

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's task taxonomy is restricted to 8 prompts spanning only two genres — explanatory and persuasive [Q38] — with the CAST rubric explicitly designed for persuasive/argumentative writing [Q118]. The deployment requires coverage of literary analysis, religious/Quranic commentary, and cultural commentary genres common in Arab high-school curricula, which the authors explicitly acknowledge as out of scope: 'limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing' [Q89]. Web research confirms this is a field-wide gap: TAQEEEM 2025 and ZaQQ 2025 maintain the same explanatory/persuasive restriction [WEB-10, WEB-11]. The benchmark also organizes its evaluation around prompt-specific and cross-prompt scoring setups [Q14, Q46], with the entire model output taxonomy restricted to numeric trait scoring [Q48, Q181] — no benchmarked task category corresponds to natural-language revision suggestion generation that the deployment requires. Given IO is HIGH priority in elicitation and multiple genre categories are missing, this represents significant taxonomic underrepresentation.

ID	Evidence
Q14	"We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt." (p.2)
Q38	"LAILA comprises 7,859 essays collected across 8 distinct prompts: 3 explanatory and 5 persuasive (3,446 and 4,413 essays, respectively)." (p.6)
Q46	"For each setup, we investigate two research questions in the context of Arabic AES: (RQ1) how do different categories of models compare in overall performance?, and (RQ2) which models achieve the best results across individual traits?." (p.7)
Q48	"The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly." (p.7)
Q89	"Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing." (p.10)
Q118	"Table 6: CAST Persuasive/Argumentative Writing Rubric - English Translation (Bashendy et al., 2024)." (p.13)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
WEB-10	TAQEEEM 2025 — first Arabic AES shared task — also limited to scoring outputs (https://aclanthology.org/2025.arabicnlp-sharedtasks.134/)
WEB-11	ZaQQ 2025 dataset also confined to standard essay genres; no Arabic AES resource for religious/literary genres (https://www.mdpi.com/2306-5729/10/9/148)

Strengths

- Within its persuasive/argumentative scope, the taxonomy is well-articulated with two complementary evaluation setups (prompt-specific and cross-prompt) that probe different generalization regimes [Q14, Q46]
- The benchmark explicitly recognizes cross-prompt evaluation as more realistic and prioritizes it for future research [Q72, Q77], aligning with deployment-style use

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Q14	"We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt." (p.2)

Q46	“For each setup, we investigate two research questions in the context of Arabic AES: (RQ1) how do different categories of models compare in overall performance?, and (RQ2) which models achieve the best results across individual traits?.” (p.7)
Q72	“We note that the applicability of the prompt-specific setup is relatively limited due to reliance on target prompt essays that are labeled, which is often unavailable in practice.” (p.9)
Q77	“Overall, the findings of the benchmarking experiments above highlight the need to prioritize future research on cross-prompt AES, which is practically closer to the real world, while best models are still far from human performance.” (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Deployment requires explanatory, persuasive, literary analysis, religious/Quranic commentary, cultural commentary, and possibly narrative/descriptive genres, plus a feedback-generation task category alongside scoring [elicitation A4; regional YAML genre_coverage].

IO-2: Yes — benchmark covers only explanatory and persuasive genres [Q38, Q89]; literary analysis, religious text commentary, and cultural commentary are entirely absent. The output task taxonomy is also limited to numeric scoring [Q48], omitting the natural-language feedback generation category that the deployment requires.

IO-3: No clear evidence that benchmark categories are irrelevant to the regional context — the persuasive and explanatory genres are themselves part of Arab curricula; the issue is omission, not inclusion of foreign categories.

IO-4: Documented gaps: (a) genre coverage missing literary, religious, cultural commentary [Q89]; (b) absence of feedback-generation task type [Q48, Q181]; (c) explanatory prompts P1 and P5 themselves showed interpretive instability with higher REL=0 frequencies [Q124, Q125], suggesting even within-scope coverage has weaknesses.

ID	Evidence
Q38	“LAILA comprises 7,859 essays collected across 8 distinct prompts: 3 explanatory and 5 persuasive (3,446 and 4,413 essays, respectively).” (p.6)
Q48	“The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly.” (p.7)
Q89	“Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing.” (p.10)
Q124	“However, P1 and P5 display relatively higher frequencies at score 0, compared to other prompts, suggesting potential ambiguity in prompt wording that may have limited clear interpretation.” (p.14)
Q125	“Notably, both P1 and P5 are explanatory prompts, which often allow greater flexibility in structure and content.” (p.14)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)
WEB-10	TAQEEM 2025 — first Arabic AES shared task — also limited to scoring outputs (https://aclanthology.org/2025.arabicnlp-sharedtasks.134/)
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Information Gaps

- Whether the deployment's specific genre mix (e.g., proportion of religious vs. literary essays) can be quantified for each target country's curriculum
- Whether explanatory prompts in LAILA approximate any deployment genre adequately enough to provide partial signal

Requires Expert Verification

- Qatari and other Arab MoE curriculum specialists should confirm the genre breakdown actually assigned at G10–12 across school types
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Remediation

Gap 1: Only explanatory and persuasive prompts are covered; literary analysis, religious/Quranic commentary, and cultural commentary genres are absent — and confirmed missing from TAQEEM 2025 and ZaQQ as well.

Recommendation: Collect a complementary Arabic essay corpus covering the missing genres at G10–12, with prompts validated against multiple national curricula, and develop genre-appropriate annotation guidelines.